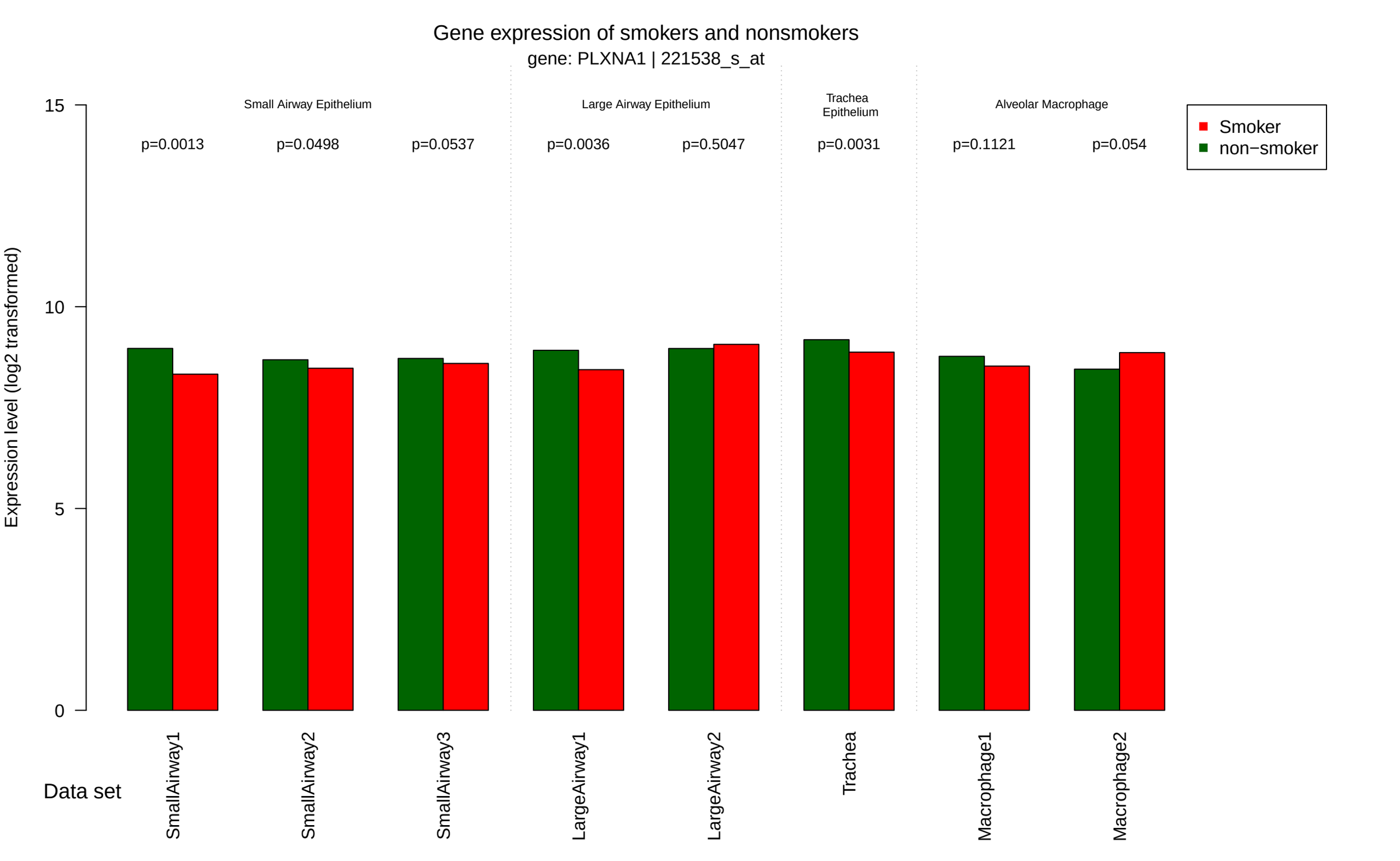
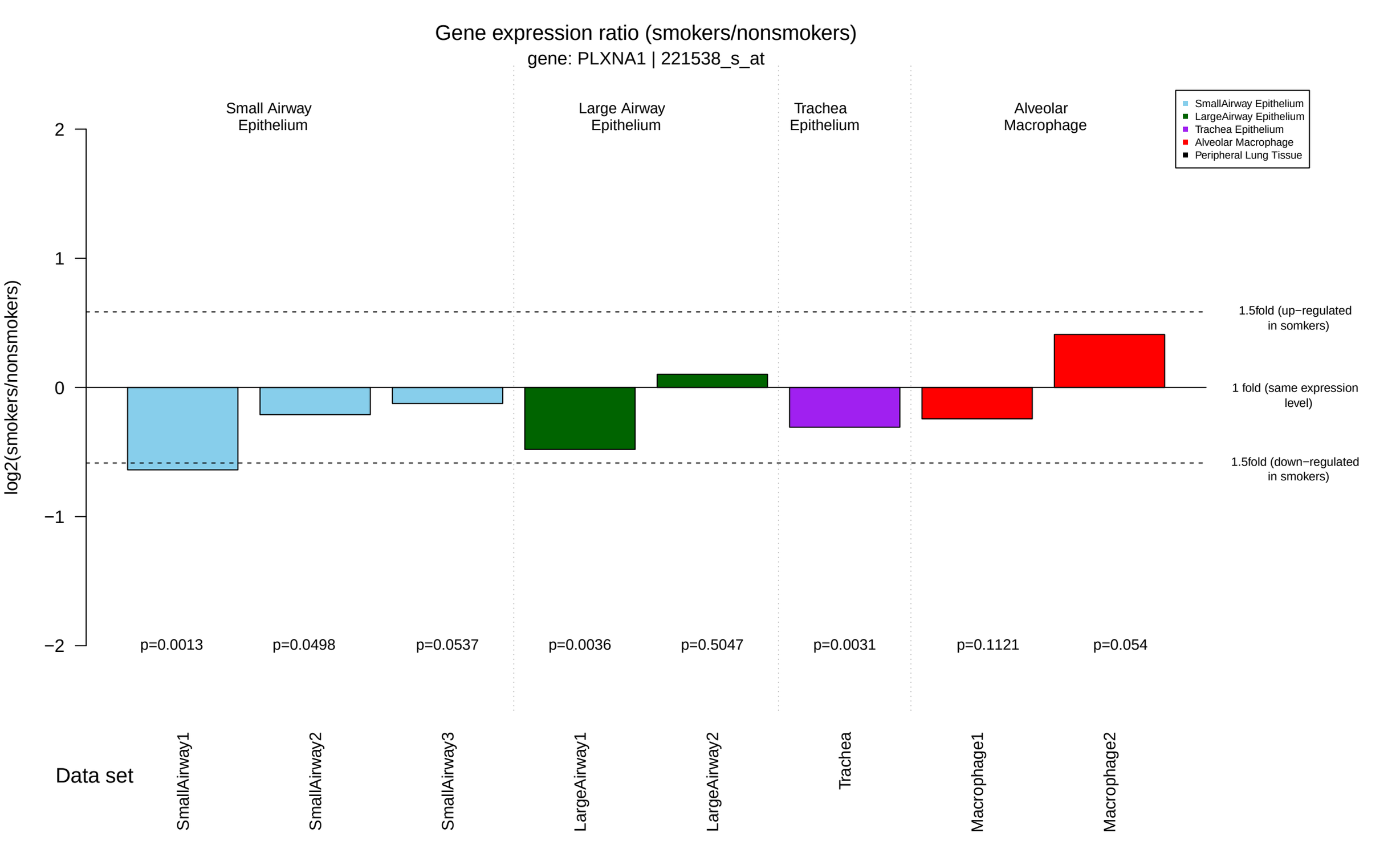
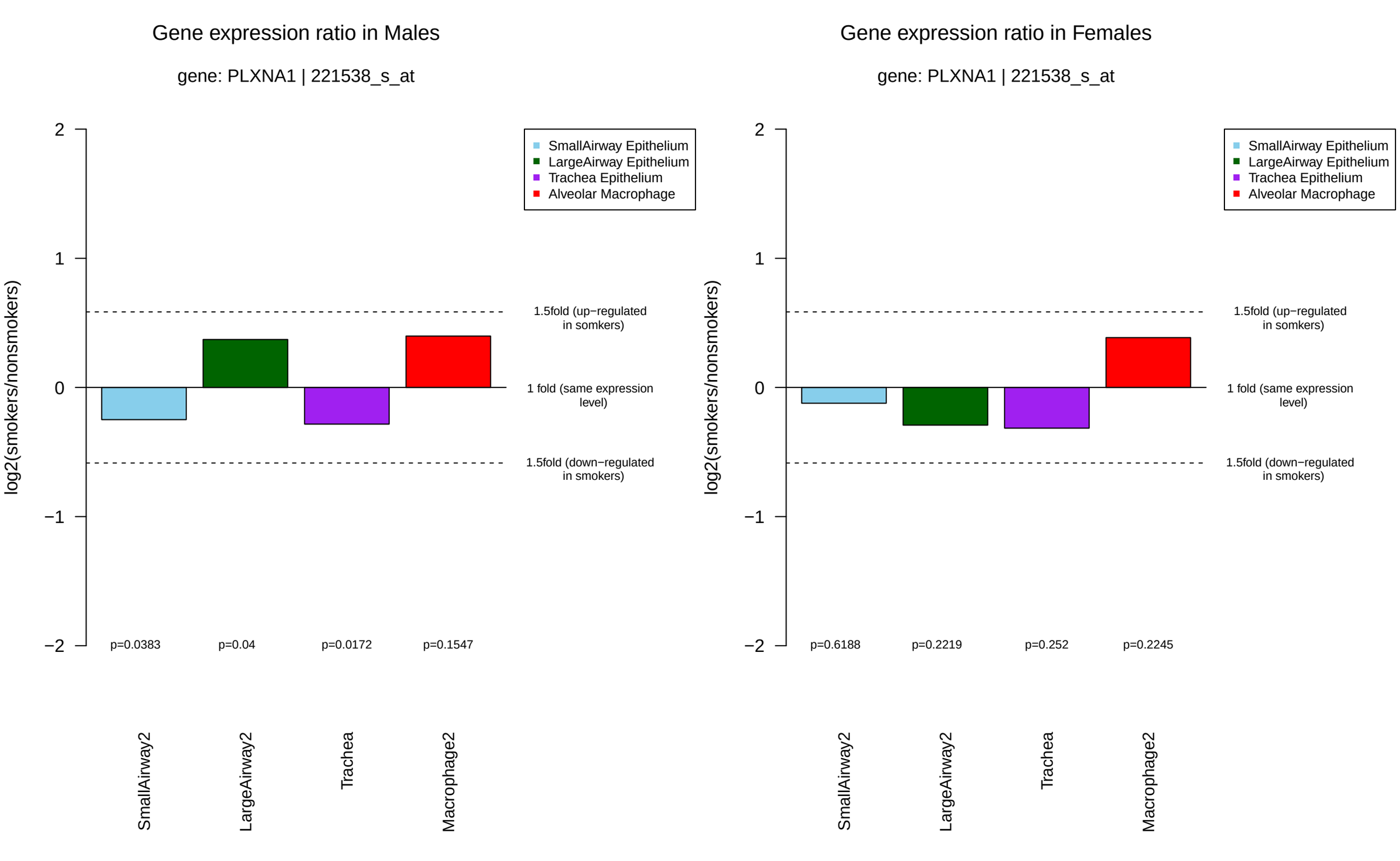
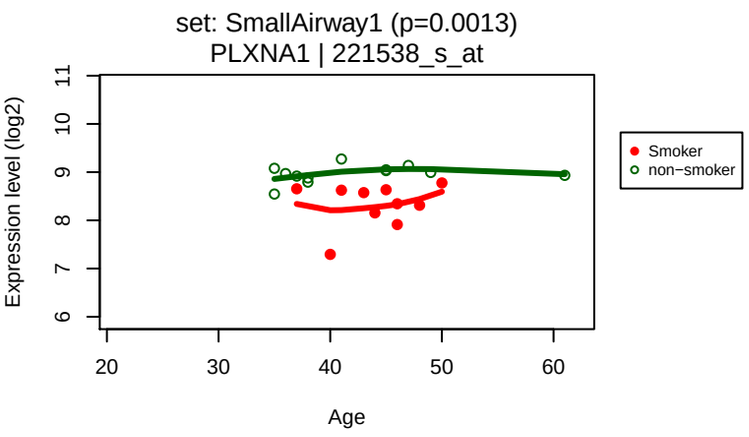
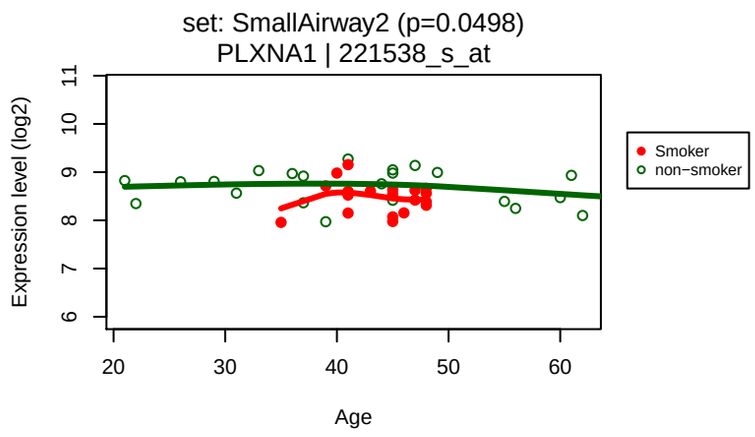
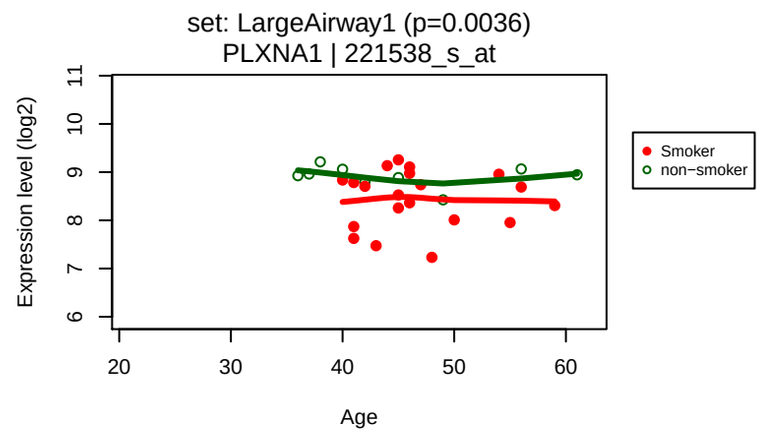

Gene expression of smokers and nonsmokers
gene: PLXNA1 | 221538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0013
p=0.0498
p=0.0537
p=0.0036
p=0.5047
p=0.0031
p=0.1121
p=0.054
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

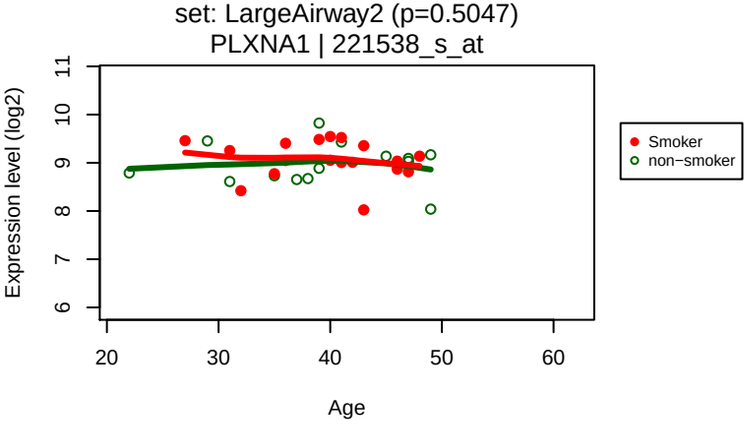

Gene expression ratio (smokers/nonsmokers)
gene: PLXNA1 | 221538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0013
p=0.0498
p=0.0537
p=0.0036
p=0.5047
p=0.0031
p=0.1121
p=0.054
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

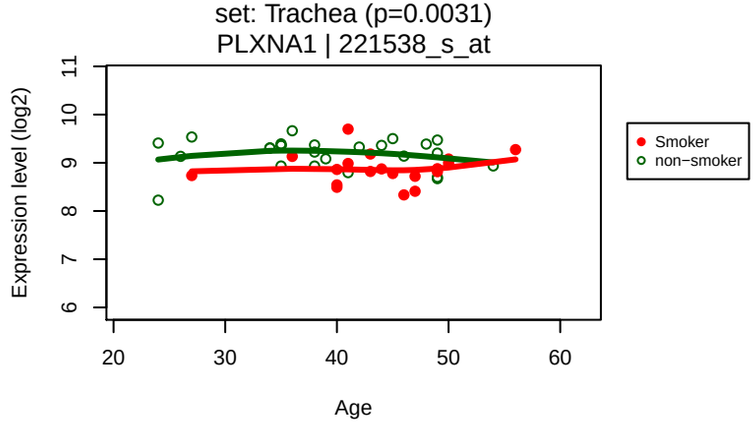

Gene expression ratio in Males
gene: PLXNA1 | 221538_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0383
p=0.04
p=0.0172
p=0.1547
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLXNA1 | 221538_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6188
p=0.2219
p=0.252
p=0.2245
SmallAirway2
LargeAirway2
Trachea
Macrophage2

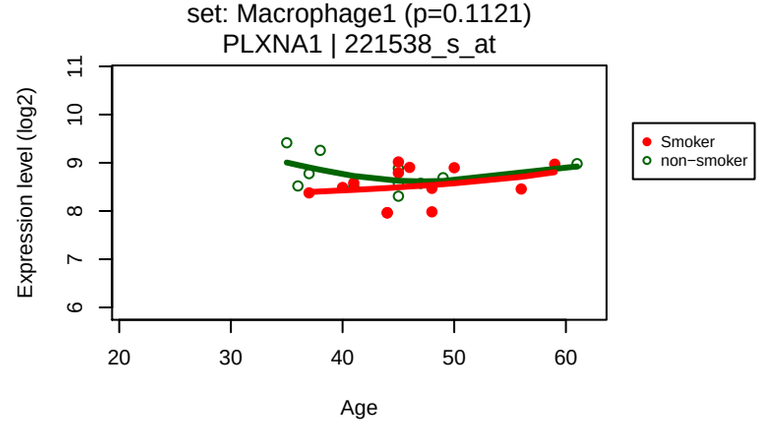

set: SmallAirway1 (p=0.0013)
PLXNA1 | 221538_s_at
Expression level (log2)
Age
Smoker
non−smoker

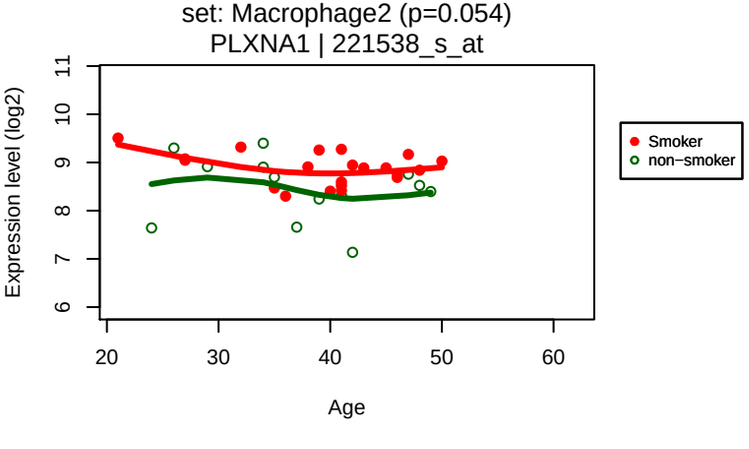

set: SmallAirway2 (p=0.0498)
PLXNA1 | 221538_s_at
Expression level (log2)
Age
Smoker
non−smoker

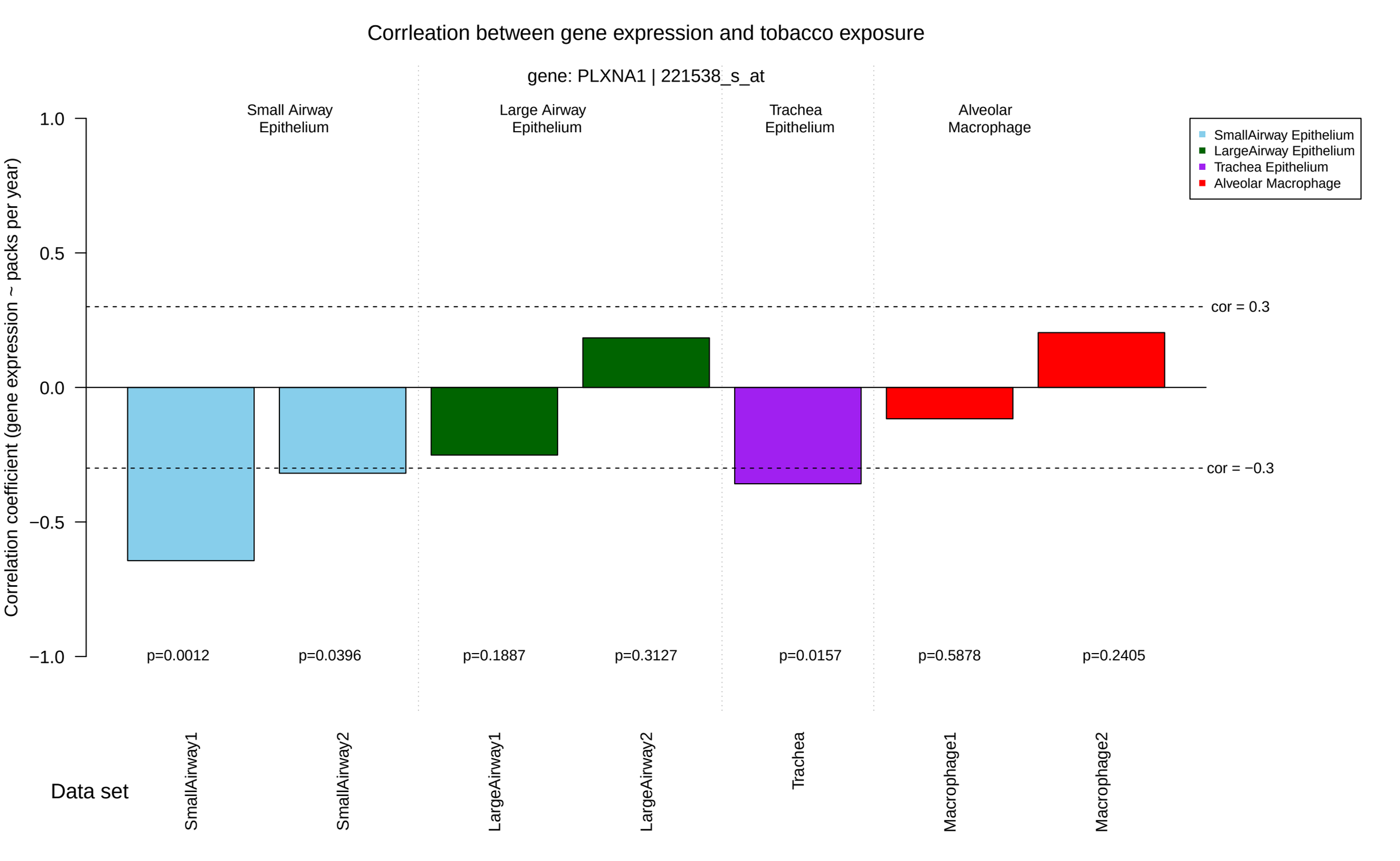

set: LargeAirway1 (p=0.0036)
PLXNA1 | 221538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5047)
PLXNA1 | 221538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0031)
PLXNA1 | 221538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1121)
PLXNA1 | 221538_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.054)
PLXNA1 | 221538_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLXNA1 | 221538_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0012
p=0.0396
p=0.1887
p=0.3127
p=0.0157
p=0.5878
p=0.2405
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2